Twenty-two counts.
More than one tuning.

A saved experiment from Grāma. The count rows and lowering instruction come from chapter 28 of the *Nāṭyaśāstra*, attributed to Bharata. The whole-gap sizes, cents and optional pure ratios are modern assumptions. This is not an ancient performance reconstruction.

Sa = 220 Hz in both witnesses. Fourth/fifth lock: on for your candidate.

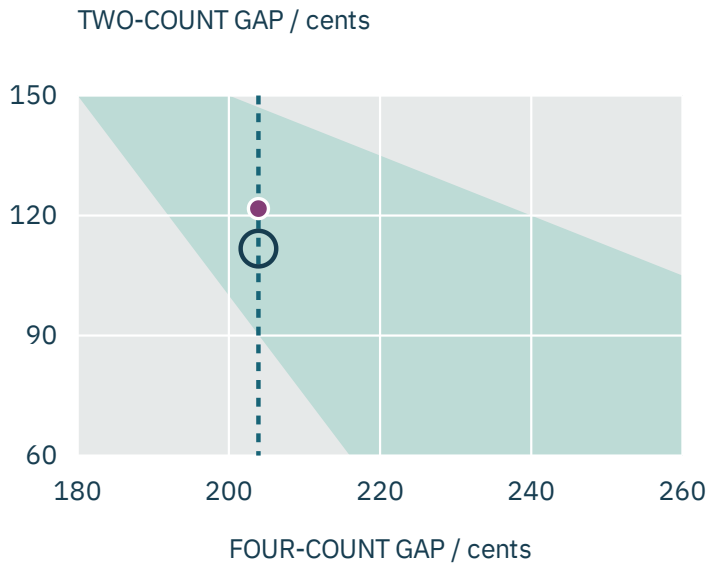
Saved witness: four = 203.910002, three = 182.403712, two = 111.731285 cents; Pa lowers by 21.506290 cents.

Your candidate: four = 203.910002, three = 172.403712, two = 121.731285 cents; Pa lowers by 31.506290 cents.

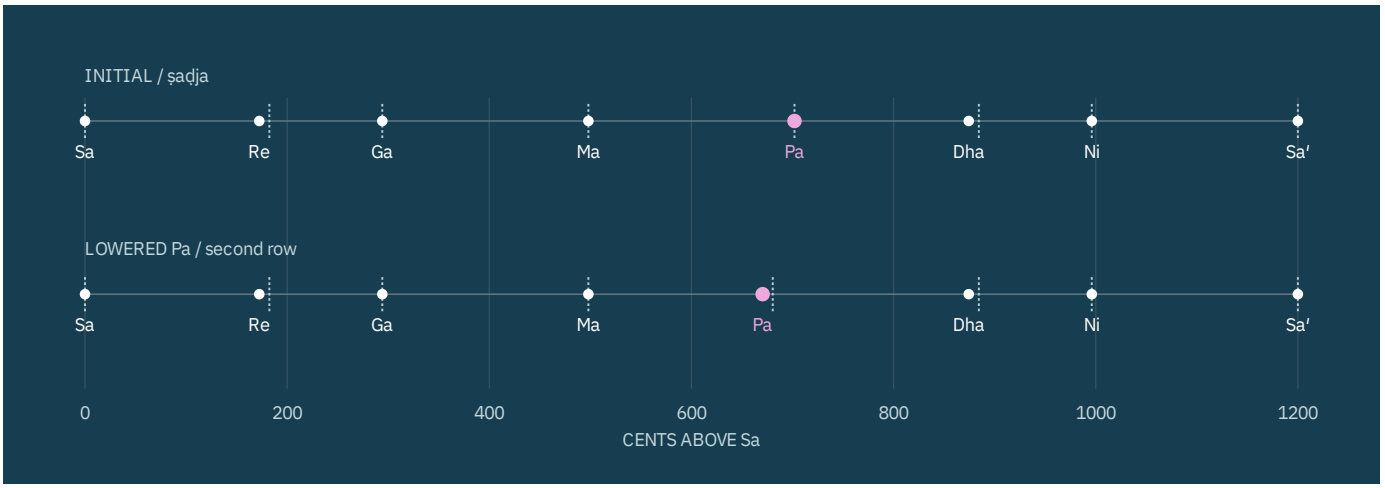
Your candidate meets the displayed count-row conditions. Pa-lowering test: passes. Ordered whole-gap test: passes. Both rows close at 1,200 cents by construction.

Two free choices, one octave equation

Let A be the whole 4-count gap, B the 3-count gap, C the 2-count gap. The ascending count row is 3–2–4–4–3–2–4; therefore 3A + 2B + 2C = 1,200 cents and B = (1,200 − 3A − 2C)/2. Repeated groups of the same count are assumed to share a size. No internal śruti pitches are assigned. Ordered sizes A > B > C > 0 are an extra condition of this model.



Shaded region: ordered whole gaps. Ring: saved witness. Filled point: current candidate.



Circles: your candidate. Dashed ticks: saved witness. Rows: initial and changed Pa. Six notes stay fixed inside each witness.

Both witnesses, same Sa = 220 Hz. Initial → second-row frequency. Six notes keep their frequencies within each witness; the witnesses can differ from each other.

Note	Saved witness / Hz	Your tuning / Hz	Your initial / cents
Sa	220.000 → 220.000	220.000 → 220.000	0.000
Re	244.444 → 244.444	243.037 → 243.037	172.404
Ga	260.741 → 260.741	260.741 → 260.741	294.135
Ma	293.333 → 293.333	293.333 → 293.333	498.045
Pa	330.000 → 325.926	330.000 → 324.049	701.955
Dha	366.667 → 366.667	364.555 → 364.555	874.359
Ni	391.111 → 391.111	391.111 → 391.111	996.090
Sa′	440.000 → 440.000	440.000 → 440.000	1200.000

Why the pure fourth and fifth do not finish the problem

In the initial row, Sa–Ma = A + B + C = 600 − A/2; Sa–Pa = 2A + B + C = 600 + A/2. Fixing these at 4/3 and 3/2 fixes A at 1,200 $\log_2(9/8)$, but leaves C free while B compensates. Pa's change is A − B. It still varies. These two ratio conditions do not represent the text's complete consonance theory.

Listen at one reference pitch

First the saved witness, then your candidate. Each plays Ma–Pa–Dha in the initial row, then in the changed-Pa row. Sine synthesis is a modern teaching aid. If the candidate fails to lower Pa, that failed

operation remains audible.

Download both listening examples

▶ Both witnesses as CSV

Your revealed listening record

Self-guided practice with immediate feedback and unlimited replay. This descriptive record does not measure a hearing threshold or establish an ancient tuning. Counts made in this tab record finished digital playback, not physical sound delivery. Reopened counts and answers come from files and are not verified here. Each attempt identifies its origin. Unfinished trials are omitted. The diagrams and editable tuning link above describe the last revealed attempt; each frozen pair is specified below.

Download the complete attempt CSV · Download the reopenable listening record

Attempt 1: Matched: you chose A; X repeats A.

Lowered Pa. Sa = 220 Hz. Largest A/B pitch difference: 10.000000 cents.

Record origin: reopened file. Past playback counts and answers are not verified in this tab.

B's count-row checks: lower Pa passes; ordered whole gaps passes. Both octaves close by construction. These are the displayed model conditions.

Frozen post-retuning pitches; rounded display

Note	A, Hz	B, Hz	B − A, cents
Pa	325.925926	324.048731	-10.000000

A whole gaps: four = 203.91000173077484, two = 111.73128526977774 cents. B whole gaps: four = 203.91000173077484, two = 121.73128526977774 cents. B fourth/fifth lock: on. Each three-count gap = (1,200 − 3 × four − 2 × two)/2. Completed playbacks: A 1, B 1, X 1. X uses exactly A's sound.

Reopen this frozen tuning pair

A (saved witness)

Download A WAV

B (candidate)

Download B WAV

The count rows alone do not choose these frequencies. The Sanskrit and full printed evidence follow in the reading.

Sanskrit and new English

Grosset's 1888 Roman Sanskrit is normalized to IAST; Devanagari renders that reading. The final half-verse definition comes from Śivadatta and Parab's 1894 edition. The editions' different verse numbering is preserved.

Two grāmas

Grosset p.28, prose before verse 25

अथ ग्रामौ । षड्जग्रामो मध्यमग्रामश्चेति । तत्र वा द्वाविंशतिश्रुतयः ।

atha grāmau. ṣaḍjagrāmo madhyamagrāmaś ceti. tatra vā dvāviṃśatiśrutayaḥ.

Now the two grāmas: the ṣaḍja-grāma and the madhyama-grāma. In them, twenty-two śrutis.

Grāmau is a dual: two grāmas. A grāma is a particular arrangement of notes and intervals. The text's claim of twenty-two is a count; this sentence does not assign a frequency ratio to each śruti. The short syllables Sa, Re, Ga, Ma, Pa, Dha, Ni are our display abbreviations.

ग्रामौ · grāmau
: the two grāmas; dual of grāma. Notice the ending -au for a pair.

श्रुति · śruti
: a heard interval distinction; the unit counted here. Do not silently replace this word with an equal division of an octave.

द्वाविंशति · dvāviṃśati
: twenty-two. The total count in either arrangement.

Three, two, four...

Grosset p.28, verse 25

तिस्रो द्वे च चतस्रश्च चतस्रस्तिस्र एव च ।
द्वे चतस्रश्च षड्जाख्ये ग्रामे श्रुतिनिदर्शनम् ॥

tisro dve ca catasraś ca catasras tisra eva ca |
dve catasraś ca ṣaḍjākhye grāme śrutinidarśanam ||

Three and two and four, and four, three as well; two and four: the indication of the śrutis in the grāma named ṣaḍja.

Read as ascending gaps beginning at Sa: Sa–Re 3, Re–Ga 2, Ga–Ma 4, Ma–Pa 4, Pa–Dha 3, Dha–Ni 2, Ni–Sa 4. The following named-note verses assign Sa four because that is the interval arriving at Sa. Starting the ascending gap row with four would put the counts on the wrong intervals.

तिस्रः · tisraḥ

three; feminine form. Before the following sound, tisraḥ becomes tisro or tisras in the verse.

द्वे · dve

two. A dual number form.

चतस्रः · catasraḥ

four; feminine form. Appears as catasraś and catasras through sandhi.

Lower Pa

Grosset p.28, prose after verse 25

मध्यमग्रामे तु श्रुत्यपकृष्टः पञ्चमः कार्यः ।

madhyamagrāme tu śrutyapakṛṣṭaḥ pañcamaḥ kāryaḥ.

But in the madhyama-grāma, Pa is to be made lower by a śruti.

Kāryaḥ is an instruction: 'to be made'. The count verses below tell us which adjacent groups change. Pa's arriving interval has three; Dha's has four. Our operation keeps the other six notes fixed. The longer two-vīṇā demonstration that follows is bracketed by Grosset as doubtful; we do not claim to reconstruct that whole experiment.

तु · tu

but; marking a contrast. The word that signals the change to the other grāma.

अपकृष्टः · apakṛṣṭaḥ

drawn down, lowered. In śrutyapakṛṣṭaḥ, the lowering is by a śruti.

पञ्चम · pañcama

fifth; the note abbreviated Pa. The one note that moves in this bench's retuning.

कार्यः · kāryaḥ

to be made or done. A verbal adjective of obligation, not a statement that it has already happened.

The first four named counts

Grosset p.29, verse 26; 1894 p.302, verse 2

षड्जश्चतुःश्रुतिर्ज्ञेय ऋषभस्त्रिश्रुतिस्तथा ।
द्विश्रुतिश्चैव गान्धारो मध्यमश्च चतुःश्रुतिः ॥

ṣaḍjaś catuḥśrutir jñeya ṛṣabhas triśrutis tathā |
dviśrutiś caiva gāndhāro madhyamaś ca catuḥśrutiḥ ||

Sa is to be known as having four śrutis, and Re three; Ga has two, and Ma four.

These are counts assigned to the named notes, read here as the preceding interval. Catuḥ-, tri- and dvi- mean four, three and two. Sandhi changes the endings of words when they meet: jñeyaḥ + ṛṣabhaḥ appears here as jñeya ṛṣabhas before the next word.

ज्ञेयः · jñeyaḥ

to be known. The instruction introducing a classification.

चतुःश्रुति · catuḥśruti

having four śrutis. A count attached to a named note.

त्रिश्रुति · triśruti

having three śrutis. Compare tri- with English triple.

द्विश्रुति · dviśruti

having two śrutis. Compare dvi- with English dual; the count remains two in both arrangements for Ga and Ni.

Pa, Dha and Ni

Grosset p.29, verse 27; 1894 p.302, verse 3

चतुःश्रुतिः पञ्चमः स्याद्धैवतस्त्रिश्रुतिस्तथा ।
निषादो द्विश्रुतिश्चैव षड्जग्रामे भवन्ति हि ॥

catuḥśrutiḥ pañcamaḥ syād dhaivatas triśrutis tathā |
niṣādo dviśrutiś caiva ṣaḍjagrāme bhavanti hi ||

Pa would have four śrutis, and Dha three; Ni has two. So they are in the ṣaḍja-grāma.

Together with the previous verse: Sa 4, Re 3, Ga 2, Ma 4, Pa 4, Dha 3, Ni 2. There are three groups of four, two of three and two of two: 12 + 6 + 4 = 22.

पञ्चम · pañcama

fifth; the note abbreviated Pa. The one note that moves in this bench's retuning.

त्रिश्रुति · triśruti

having three śrutis. Compare tri- with English triple.

द्विश्रुति · dviśruti

having two śrutis. Compare dvi- with English dual; the count remains two in both arrangements for Ga and Ni.

The two changed intervals

Grosset p.29, verse 28

चतुःश्रुतिस्तु विज्ञेयो मध्यमः पञ्चमः पुनः ।
त्रिश्रुतिर्धैवतस्तु स्याच्चतुःश्रुतिक एव हि ॥

catuḥśrutis tu vijñeyo madhyamaḥ pañcamaḥ punaḥ |
triśrutir dhaivatas tu syāc catuḥśrutika eva hi ||

Ma is to be known as having four śrutis; Pa, in turn, three. But Dha would have four.

The text now begins its list at Ma. Keep the note names attached to their counts: Ma 4, Pa 3, Dha 4. Lowering Pa contracts Ma–Pa and expands Pa–Dha; their combined seven counts remain seven.

तु · tu
: but; marking a contrast. The word that signals the change to the other grāma.

पञ्चम · pañcama
: fifth; the note abbreviated Pa. The one note that moves in this bench's retuning.

त्रिश्रुति · triśruti
: having three śrutis. Compare tri- with English triple.

चतुःश्रुति · catuḥśruti
: having four śrutis. A count attached to a named note.

The remaining notes

Grosset p.29, verse 29

निषादषड्जौ विज्ञेयौ द्विचतुःश्रुतिसंभवौ ।
ऋषभस्त्रिश्रुतिश्च स्याद्गान्धारो द्विश्रुतिस्तथा ॥

niṣādaṣaḍjau vijñeyau dvicatuḥśrutisambhavau |
ṛṣabhas triśrutiś ca syād gāndhāro dviśrutis tathā ||

Ni and Sa are to be known as having two and four śrutis respectively; Re would have three, and Ga two.

The dual niṣādaṣaḍjau joins the two note names. With the previous verse, this gives the full second arrangement. Our display keeps Sa at the left in both rows so that moving Pa is visible; the name madhyama-grāma does not force the player to begin every ascent on Ma.

द्विश्रुति · dviśruti
: having two śrutis. Compare dvi- with English dual; the count remains two in both arrangements for Ga and Ni.

त्रिश्रुति · triśruti
: having three śrutis. Compare tri- with English triple.

चतुःश्रुति · catuḥśruti
: having four śrutis. A count attached to a named note.

Seven notes in succession

1894 p.302, first half of verse 8

क्रमयुक्ताः स्वराः सप्त मूर्छनेत्यभिसंज्ञिताः ।

kramayuktāḥ svarāḥ sapta mūrchanetyabhisaṃjñitāḥ |

Seven notes joined in succession are designated a mūrchanā.

This is the first half of the verse, not its entire text. Changing the first note gives a different ordered ascent through an existing tuning. Our seven ascending rotations illustrate this structural distinction; they do not reproduce a historical melody, ornamentation or the full theory of named mūrchanās.

क्रम · krama

succession; order. The order is changed when you choose a new starting note.

स्वर · svara

note; musical tone. Svarāḥ is plural: notes.

सप्त · sapta

seven. Seven named notes, with the starting note repeated at the octave in our ascent.

मूर्छना · mūrchanā

an ordered ascending or descending succession. We use ascending rotations to distinguish order from retuning. Historical performance is richer than this model.

Where this experiment stops

Meeting the displayed checks does not establish a historically used tuning, equal sizes for individual śrutis, or satisfaction of all additional ancient interval relations. Grosset brackets the longer two-vīṇā passage on pp.28–29; note 36 on p.86 describes his interpretation as approximate. This family does not resolve that passage. The text is composite; no exact date is assigned to it here. All displayed cents and Hz are numerical approximations.

Full printed evidence

Joanny Grosset, *Contribution à l'étude de la musique hindoue* (1888); Śivadatta and Kāśīnāth Pāṇḍuraṅg Parab, eds., *The Nāṭyaśāstra of Bharata Muni*, Kāvyamālā 42 (1894). Full edition and scan identities are retained below.

158 A 81 (2)

CONTRIBUTION A L'ETUDE

DE

LA MUSIQUE HINDOUE

PAR

J. GROSSET

BOURSIER D'ÉTUDES PRÈS LA FACULTÉ DES LETTRES DE LYON

grosset1888, printed p.1, physical PDF p.163. Title of Grosset's contribution
SHA-256 f900e5c8c737b1ac3e99af0fb83e161ebb2a7781e48ed94ee7298f0f4c9a1bb3

et de vers *dryas*. L'état des deux manuscrits laisse beaucoup à désirer : les variantes sont nombreuses, les leçons diffèrent sensiblement en plus d'un endroit.

Le ms. G présente, en dehors de nombreuses divergences de détail, des déplacements importants coupés de lacunes, comparativement au texte de A. L'ordre des *çlokas* est interverti nôtamment du *çl.* 11 au *çl.* 21, du *çl.* 48 au *çl.* 54; enfin, à partir du *çloka* 78, le désordre est tel que nous avons dû, faute de pouvoir indiquer clairement en note les déplacements et les rapprochements, donner à part, jusqu'à la fin de l'*adhyâya*, le texte de G en appendice.

Nous constatons en outre des intercalations évidentes et des lacunes. Parmi les passages intercalés, nous signalerons les 4 *çlokas* 108-111 empruntés à A et qui manquent dans G. Rien n'annonce le développement en question, qui se termine, du reste, par un fragment de *pada*, suivi d'une lacune. Nous ferons la même observation pour les *çlokas* 39-40, pris cette fois dans G et qui font défaut dans A. Nous avons encore des raisons de croire à une lacune après le *çloka* 80 qui ne satisfait pas au développement annoncé.

Certaines répétitions de mots, de phrases entières, sont dues à l'inattention du copiste (par ex. dans A : les 4 hémistiches intercalés à l'intérieur du *çloka* 45, les quelques lignes de prose entre *çl.* 81 et 82; l'hém. 144 *a* répété après 145. — Dans G : les deux hém. 91 *b*, 92 *a*, les 3 hém. 91-93 intercalés *çl.* 134, et qui reviennent plus loin régulièrement, fondus en 2 hém.,... etc.).

En plus d'un endroit, le mauvais état des mss., qui gardent des traces de remaniements évidents, rendent peu sûrs l'établissement du texte et son interprétation. Nous avons pris comme règle de mettre entre crochets les passages douteux. Nous avons dû, en outre, hasarder certaines restitutions : nous les avons réduites au strict nécessaire, et nous y avons procédé avec la plus grande réserve. En général, nous étions autorisé à les faire par les variantes qu'offrait l'un ou l'autre de nos mss.; c'est parfois la grammaire, le

grosset1888, printed p.21, physical PDF p.183. Editorial brackets explained
SHA-256 79f6b71fbce81f18d63d663ce10883a2a17cc15fe2adb76b824acf426148a309

1 saṃjñakâḥ.
yathâ. ṣaḍjasyarṣabhagâṇdhâradhaivataniṣâdâḥ. ṛṣabhasyamadhyamapañcamaniṣâdâḥ. gâṇdhârasyâpi madhyamapañcamadhaivatâḥ. madhyamasya dhaivatapañcamaniṣâdâḥ.pañcamasya dhaivatašaḍjau. dhaivatasya šaḍjamadhyamapañcamâḥ šaḍjagrâmo.
madhyamagrâmo 'pi madhyamasya pañcamadhaivataniṣâdâḥ. pañcamasyarṣabhašaḍjagâṇdhârâḥ. dhaivatasya šaḍjarṣabhagâṇdhârâḥ. niṣâdasya šaḍjarṣabhagâṇdhârâḥ.
vadanâd vâdî saṃvadanât saṃvâdî vivadanâd vivâdy anuvadanâd anuvâdîti.
eteṣâṃ ca svarâṇâṃ nyûnâdhikatvaṃ tantrîvâdanadaṇḍendriyavaiguṇyâd upajâyate.
svaravidhânam etac caturḥprakâram iti.
atha grâmau.
ṣaḍjagrâmo madhyamagrâmaç ceti.
tatra vâ dvâviṃçatiçrutayaḥ. yathâ.
tisro dvo ca catasraç ca catasras tisra eva ca |
dve catasraç ca šaḍjâkhyo grâmo çrutinidarçanam || 25 ||
madhyamagrâmo tu çrutyapakṛṣṭaḥ pañcamaḥ kâryaḥ.
pañcamaçrutyutkarṣâd apakarṣâd vâ yad antaraṃ mârdavâd âyatatvâd vâ tat pramâṇaçrutiḥ.
nidarçanaṃ tv âsâm abhivyâkhyâsyâmaḥ. yathâ.
[dve vîṇe tulyapramâṇatantryupavâdanadaṇḍamârohaṇe
ṣaḍjagrâmâçrite kârye. tayor ekatarasyâṃ madhyamagrâmikîṃ kṛtvâ pañcamasyâpakarṣe çrutiṃ tâm eva pañca-

2-9 m. dans A. šaḍjasya ṛšaº G. — 3 madhyapañcº G. — 5 ºšodau G. — 8 pañcamasya omis par G. — 9 šaḍjaº G. — 10 vâdanât vâdî saṃvâdânât saṃvâdîr vivâditvât vivâdî G. vivâdîti A. — 11 anuvâdanâd ânuvâdî G. omis dans A. — 12 ca manque dans G. ûnâº G. ºdhikatva A. tantryupaº G. vâdanaṃ A. — 13 ºguṇyad G. — 14 ity etat svaravirṇadhânam caturvidhâḥ G. — 15 et 16 atha dvau grâmau grâmau šaḍjo madhyamaç ceti G. — 17 nirâçcintâ A. tatrâ vâ G. — 18 ca omis après dve G. — 19 ºkhyo A. tu çrutidaiº G. — 20 çrutyacatuṣṭhuḥ G. — 21 pañcamasya çrutyutkarṣâbhyâm yad antaraṃ G. anantaraṃ A. — 22 mârdavâyatatvam tâvat pramâṇaçruti G. pramâṇâº A. — 23 câsâm abhikhyâº G. — 24 vîṇo A. ºvâdanam A. daṇḍa manque dans G. — 25 et 26 mûrcchagrâmo šaḍjaº G. ºgramâº A. tayor anyatarâmadhyamanâmikîṃ kuryât G. ºkarṣa G. çrutiḥ | A.

grosset1888, printed p.28, physical PDF p.190. Two grāmas, counts and lowering Pa
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mavaçât šaḍjagrâmikam kuryât. evaṃ çrutir apakṛšṭâ bhavati.
punar api tadvad evâpakaršâd gândhâranišâdavantâv itarasyâṃ dhaivataršabhau praviçato dviçrutyadhikatvât.
punas tadvad evâpakaršâd dhaivataršabhâv itarasyâṃ pañcamašaḍjau praviçataḥ çrutyadhikatvât.
tadvat punar apakṛšṭâyâṃ tasyâṃ pañcamamadhyamašaḍjâ itarasyâṃ madhyamagândhâranišâdavantaḥ pravekšyanti catuḥçrutyadhikatvât.]
evam anena çrutinidarçanena dvaigrâmikyo dvâvimçatiçrutayaḥ pratyavagantavyâḥ.
atra çlokâ bhavanti.
šaḍjaç catuḥçrutir jñeya ṛšabhas triçrutis tathâ |
dviçrutiç caiva gândhâro madhyamaç ca catuḥçrutiḥ || 26 ||
catuḥçrutiḥ pañcamaḥ syâd dhaivatas triçrutis tathâ |
nišâdo dviçrutiç caiva šaḍjagrâme bhavanti hi || 27 ||
catuḥçrutis tu vijñeyo madhyamaḥ pañcamaḥ punaḥ |
triçrutir dhaivatas tu syât catuḥçrutika eva hi || 28 ||
nišâdašaḍjau vijñeyau dvicatuḥçrutisaṃbhavau |
ṛšabhas triçrutiç ca syâd gândhâro dviçrutis tathâ || 29 ||
antaranidarçanam api vyâkhyâtam.
atha mûrchanâḥ.
dvaigrâmikyaç caturdaça. yathâ.
âdâv uttaramandrâ syâd rajanî cottarâyatâ |

1 pañcamasya çrutyutkaršavaçât šaḍjagrâmikam kuryât ata çrutir apakṛšṭâ.. G. — 3 evapakaršayat yathâ A. evârakaršât G. °nišâdâv api G. — 4 itarasyâṃ pravekšyanti [lacune jusqu'après pravekšyanti, I. 8] G. °ršâbho praviçataḥ dviçrutyâbhyadhiº A. — 5 °karšâdhvaivatarišabhâdh A. — 8 A coupe la phrase après pravekšyanti. — 9 catuçcatyabhyadhiº G. — 10 çrutidaçanavidhânena G. dvaugrâmiko A. dvâvimçâ çruº G. — 11 A et G coupent après çrutayaḥ | . pratyagantº G. — 12 atra çlokâḥ | (bhavanti manque) G. — 13 catuçruti G. triçrutismitaḥ G. — 14 câpi G. catuçruº G. — 15 catuçcatuḥ | G pañcama A, G. syâ triçrutir dhaivatam G. — 16 dviçrutis tu nišâda syât G. svarântaro | G. — 17-21 manquent dans A. 17 catuçruº G vyšeyâ G. — 18 catuçruº G. — 19 G écrit exceptionnellement ici šaḍjau. catuçruº G. — 20 triçrutiç ca syât G — 22 Nous coupons après mûrchanâ, en ajoutant le ḥ; 22, 23 mûrcchanâḥ.o grâmº G. dvaigrâmikâç caiva .. A. yathâ manque dans G. — 24 âdyâḥ uttaramantrâ âkhyâ rajanani... G.

grosset1888, printed p.29, physical PDF p.191. Counts attached to each named note
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Le terme *vâdin* dérive de *vadana; saṃvâdin* de *saṃvadana; vivâdin* de *vivadana; anuvâdin* de *anuvadana.*

? Le manque de justesse[31] par excès ou différence *(nyûnâdhikatva)* dans la production des sons [rangés sous ces quatre catégories] résulte de l'imperfection *(vaiṣamyât)* du *daṇḍa* [ou manche de l'instrument] et du défaut de résonnance des cordes.

Telles sont les quatre catégories des notes.

LES MODES DE LA GAMME *(grâma*[32]*).* —

Il y a deux modes de la gamme : le mode *ṣaḍja* et le mode *madhyama*[33].

LES VINGT-DEUX INTERVALLES *(çruti).* —

Chacun de ces modes comprend vingt-deux intervalles[34].

25 En voici le décompte pour le *ṣaḍjagrâma :*

trois, deux, quatre, quatre, trois, deux, quatre.

(prose) Dans le *madhyamagrâma, pañcama* doit être diminué *(apakṛṣṭa*[35]*)* d'un intervalle *(çruti).* L'intervalle d'une *çruti (antara)* qui représente l'élévation *(utkarṣa)* ou l'abaissement *(apakarṣa)* de *pañcama,* sa diminution *(mârdava)* ou son augmentation *(âyatatva),* est la *çruti*-type, indicatrice *(pramâṇa).*

EXPOSÉ DÉMONSTRATIF DE LA THÉORIE DES *çrutis*[36].

?? Soient deux luths *(vîṇâ)* accordés en *ṣaḍjagrâma,* ayant même *mûrchanâ,* même *daṇḍa,* même résonnance des cordes et même *çruti*-type. On fait dans l'un ou l'autre de ces luths la *çruti* du mode *madhyama,* en diminuant *pañcama,* de telle sorte que par une simple modification de *pañcama (pañcamavaçât)* on pourrait rétablir la *çruti* du mode *ṣaḍja :* on a ainsi diminué d'une *çruti.*

On procède à une nouvelle diminution : les notes *gândhâra* et *niṣâda* de l'un des luths deviennent alors

grosset1888, printed p.57, physical PDF p.219. French translation of counts and lowering
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dans l'autre *dhaivata* et *ṛ̀abha*, par suite de l'addition de deux *çrutis* (?).

Nouvelle diminution, en vertu de laquelle *dhaivata* et *ṛ̀abha* du premier deviennent dans le second *pañcama* et *ṣadja*, grâce à l'addition de [trois] *çrutis* (?).

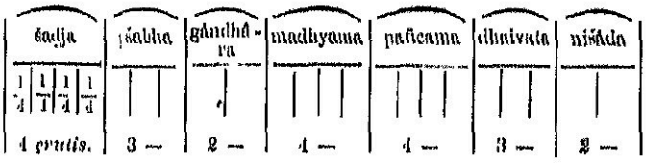
Enfin, subissant encore une diminution, le premier voit ses notes *pañcama*, *madhyama* et *ṣadja* devenir dans l'autre *madhyama*, *gándhára* et *niṣáda*, par l'addition de quatre *çrutis* (?).

— La démonstration précédente permet de saisir le détail des vingt-deux *çrutis* des deux *grámas*.

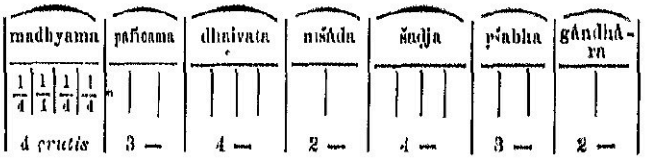
Suivent des *çlokàs* (26-29).

DISPOSITION DES *çrutis* DANS LES DEUX MODES DE LA GAMME

Mode ṣadja



Mode madhyama



(prose) Tel est le tableau des intervalles *(antara)*.

Les *Mûrchands* [37]. —

Il y a quatorze *mûrchanâs* [ou séries continues des notes successives] pour les deux modes [réunis].

grosset1888, printed p.58, physical PDF p.220. Count diagrams and mūrchanās
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pas cru pouvoir, sans y être autrement autorisé, hasarder cette restitution. Disons cependant que tous les théoriciens s'accordent à mentionner les notes *ri-ga*, *dha-ni* comme dissonantes.

30 *Notes auxiliaires.* La définition est présentée sous une forme peu satisfaisante. Il n'était pas besoin d'indiquer qu'une même note ne pouvait, étant tonique, être en même temps sa propre auxiliaire. Pour dresser le tableau, nous n'avons à notre disposition que le texte de G, très sujet à caution. Nous le donnons cependant tel quel, mais en remarquant que dans ce tableau figurent d'une part *pa-sa*, de l'autre *pa-ri*, *ni-ga* qui, d'après les indications précédentes, sont déjà consonantes, — tout comme les notes *dha-ri* pour lesquelles nous avons fait des réserves.

31 *Manque de justesse.* Nous n'osons répondre que cette interprétation soit exacte ; mais nous n'en voyons pas d'autre possible.

32 *grâma.* Le mot *grâma*, littéralement : groupement, village, signifie, dans son acception musicale, un groupe (*samûha*, *saṃdoha*) de notes disposées convenablement (*suvyavasthâna*). Le prof. A. Weber (*Ind. Lit. Geschichte*, 2e éd., p. 367-8), croit pouvoir faire dériver de ce mot sanskrit *grâma* (prâkrit *gama*) le français *gamme* et l'anglais *gamut*, empruntés au *gamma* de Gui d'Arezzo ; et y voir un témoignage direct de l'origine hindoue de notre échelle européenne à sept notes. (Comparez *Public Opinion*,... publié par Mohun Tagore, supplément, p. 12 ; et A. Weber, *Ind. Streifen*, III, p. 544.)

33. *Mode ṣaḍja, madhyamâ.* Nous n'avons pas cru devoir, pour éviter toute confusion, traduire, conformément au procédé habituel, par *gamme ṣaḍja*, *gamme madhyama*. Les deux *grâmas* sont, selon nous, deux modes distincts de la gamme ; il y a entre eux une différence analogue à celle qui sépare, dans notre musique occidentale, le mode majeur du mode mineur.

Dans le *madhyamagrâma*, — du moins d'après notre texte, — la série des notes commençant à *ma* (Mohun Tagore la fait commencer à *sa*), ne présente pas le même nombre de *çrutis* disposées dans le même ordre que dans le *ṣaḍjagrâma*, qui peut être considéré comme le mode de la gamme-type. En d'autres termes les intervalles affectés aux notes ne suivent pas la même disposition dans les deux *grâmas* : ils diffèrent par autre chose que par le choix de la tonique. Il n'y a que deux *çrutis* en *ṣaḍja* entre la troisième et la quatrième, il y en a quatre en *madhyama* ; inversement, tandis que le premier de ces modes présente quatre *çrutis* entre la quatrième et la cinquième, le second n'en compte que deux.

Ajoutons que la différence est encore plus sensible entre les deux *grâmas* que nous venons d'indiquer et le *gândhâragrâma*, un troisième mode que plusieurs théoriciens, postérieurs à Bharata, mentionnent. Ils déclarent, il est vrai, qu'il est tombé en désuétude, ou même qu'il n'a jamais été en usage que chez les dieux, ce qui ne les empêche pas, du reste, d'en présenter la définition et l'échelle.

34 *Vingt-deux intervalles.* L'ordre naturel de cette énumération serait : quatre, trois, deux, quatre, quatre, trois, deux. Le déplacement s'explique peut-être par les nécessités du mètre.

35 *apakṛṣṭa.* En d'autres termes, *pañcama* perd une *çruti* au profit de *dhaivata* ; et se trouve réduit à ses trois premières *çrutis* (*svapâdâçrutisam-*

grosset1888, printed p.85, physical PDF p.247. Commentary on the two grāmas
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sha) en *madhyama*, alors que, dans la gamme primitive, cette même note *pañcama* est naturelle (*nirvikârin*). — (*Sangîta-sâra-sangraha*, p. 28).

Cette théorie n'est pas celle que présente Mohun Tagore. Pour lui « le *madhyamagrâma* est déduit du *Sadja* par la réduction de l'intervalle entre *pa* et *dha* d'une *çruti*, et la position de *ni* deux *çrutis* plus bas dans l'échelle (*Six principal Râgas*, Introduction, p. 24).

36 *Théorie des çrutis.* Nous donnons, en suivant pas à pas le texte, une traduction approximative de cette démonstration obscure, dont nous n'avons pu dégager un sens satisfaisant. La lacune du ms. G, résultat d'une confusion évidente, contribue encore à rendre douteuse la lecture du seul manuscrit que nous ayons à notre disposition.

Nous en sommes réduit à remarquer que les deux notes du premier luth, dont il est question tout d'abord, *ga* et *ni*, possèdent chacune 2 *çrutis* (2×2 = 4); les deux suivantes, *ri* et *dha*, chacune 3 (3×2 = 6); les trois dernières *pa*, *ma*, *sa*, chacune 4 (4×3 = 12), ce qui nous donne bien, au total, les 22 *çrutis* de l'échelle musicale.

37 *Les mûrchanâs.* La série des notes qui constituent la gamme peut être ou ne pas être *mûrchanâ*. Quand ces notes sont considérées comme produites chacune séparément, sans liaison, il n'y a pas *mûrchanâ*. Mais si elles se suivent sans interruption de l'une à l'autre dans l'échelle ascendante ou descendante, en produisant une série continue de sons régulièrement disposés, alors les *mûrchanâs* prennent naissance. En d'autres termes, il y a *mûrchanâs* dans une gamme qu'on monte ou qu'on descend en lisant les notes, dans une seule émission de voix.

C'est du moins l'interprétation que nous suggèrent les textes sanskrits, car il est difficile d'imaginer une diversité d'opinion plus grande que celle qui a divisé sur ce point les divers auteurs européens.

Les définitions des théoriciens hindous sont ordinairement d'un vague désespérant. Nous citerons seulement celle que donne le *Sangîta darpaṇa* (*Sangîta-sâra-sangraha*, p. 30); elle est un peu plus explicite :

« La montée (*âroha*) ou la descente (*avarohaṇa*) des sept notes en gradation (*kramât*), c'est ce qui constitue la *mûrchanâ* : il y en a sept pour chacun des trois *grâmas*. »

38 *uttaramandrâ*, etc. Les dénominations affectées aux quatorze *mûrchanâs* — portées à vingt-et-une par les théoriciens qui admettent trois modes, — sont empruntées aux idées et aux objets les plus divers, et ne paraissent pas mériter une grande attention.

Elles diffèrent quelque peu, suivant les auteurs, et ne sont pas toujours disposées dans le même ordre, ni affectées à la même série de notes.

Remarquons à côté de *abhiruḍgatâ* la forme *abhyuḍgatâ*, et l'analogie que présente *mârgavî* avec le nom du *sâman mârgaviya* (Burnell : *The Arsheyabrâhmaṇa*, p. 20. — Comparez pour les noms des *sâmans* : *Introduction*, p. xxxv, xxxvm).

39 *de deux manières.* Pour avoir la clef de ce passage difficile, il serait nécessaire de comprendre préalablement la démonstration de la théorie des *çrutis* à laquelle l'auteur renvoie. C'est dire que nous, nous bornons, comme plus haut, à une interprétation approximative.

40 *Les tânas.* Nous n'avons pas de définition des *tânas*. De ce que ce mot ne

grosset1888, printed p.86, physical PDF p.248. Doubt about the two-vīṇā demonstration
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KÂVYAMÂLÂ. 42.

THE
NÂTYAS'ÂSTRA
OF
BHARATA MUNI.

EDITED BY

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अथवा देशकालौ च न परीक्ष्यौ प्रयोक्तृभिः ।
यदैवाज्ञापयेद्भर्ता तदा योज्यमसंशयम् ॥ ८६ ॥
तथा समुदिताश्चैव विज्ञेया नाटकाश्रयाः ।
पात्रं प्रयोगमर्थश्च विज्ञेयास्तु त्रयो गुणाः ॥ ८७ ॥
बुद्धिः सत्त्वं सुरूपत्वं लयतालज्ञ(?)नान्यथा ।
रसभावज्ञता चैव वयःस्वत्त्वकुतूहलम् ॥ ८८ ॥
ग्रहणं धारणं चैव गात्रवैकल्यमेव च ।
जितसाध्वसतोत्साह इति **पात्र**गतो विधिः ॥ ८९ ॥
सुचये(चेय)ता सुगानत्वं सुपाठ्यत्वं तथैव च ।
शास्त्रकर्मसमायोगः **प्रयोगः** संप्रकीर्तितः ॥ ९० ॥
सुविभूषणता या तु सुमाल्याम्बरता तथा ।
विचित्रा रचना चैव **समृद्धि**रिति सा स्मृता ॥ ९१ ॥
यदा समुदयाः सर्वे एकीभूता भवन्ति हि ।
अलंकारः स तु तदा मन्तव्यो नाटकाश्रयः ॥ ९२ ॥
एतदुक्तं द्विजश्रेष्ठाः सिद्धीनां लक्षणं मया ।
अत ऊर्ध्वं प्रवक्ष्याम्यातोद्यानां च विकल्पनम् ॥ ९३ ॥
इति भारतीये नाट्यशास्त्रे सिद्धिव्यञ्जको नामाध्यायः सप्तविंशतितमः ।

अष्टाविंशतितमोऽध्यायः ।

अथातोद्यविधिमिदानीं वक्ष्यामः—

ततं चैवावनद्धं च घनं शुषिरमेव च ।
चतुर्वि[धं] ज्ञेयमस्यापकर्षे श्रुति(त)मेव च ॥ १ ॥

पञ्चमवशात्षड्जग्रामिकीं कुर्यात् । एवं श्रुतिरपकृष्टा भवति । पुनरपि तद्वदेवापकर्षयेत् । यथा—गान्धारनिषादवन्तावितरस्यामृषभधैवतौ प्रवेक्ष्यतो द्विश्रुत्यभ्यधिकत्वात् । तद्वत्पुनरपकृष्टायां तस्यां पञ्चममध्यमषड्जा इतरस्यां मध्यमगान्धारनिषादवन्तः प्रवेक्ष्यन्ति चतुःश्रुत्यभ्यधिकत्वात् । पवतेन श्रुतिनिदर्शनेन द्वैग्रामिक्यौ द्वाविति श्रुतयः प्रत्यवगन्तव्याः ।

natya1894, printed p.301, physical PDF p.304. Chapter opening and end of the two-vīṇā passage
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तत्र श्लोकौ भवति(तः)—

षड्जश्चतुःश्रुतिर्ज्ञेय ऋषभस्त्रिश्रुतिस्तथा ।
द्विश्रुतिश्चैव गान्धारो मध्यमश्च चतुःश्रुतिः ॥ २ ॥
चतुःश्रुतिः पञ्चमः स्याद्धैवतस्त्रिश्रुतिस्तथा ।
निषादो द्विश्रुतिश्चैव षड्जग्रामे भवन्ति हि ॥ ३ ॥

शुद्धमूर्छना द्वैग्रामिक्यश्चतुर्दश । यथा—

आदावुत्तममन्द्रा स्याद्रजनी चोत्तरायता ।
चतुर्थी शुद्धषड्जा च पञ्चमी मत्सरीकृता ॥ ४ ॥
अश्वक्रान्ता तथा षष्ठी सप्तमी चाभिरुद्गता ।
षड्जग्रामाश्रिता चैव विज्ञेयाः सप्त मूर्छनाः ॥ ५ ॥
आस(?)षड्जनिषण्णाश्च यस्यात्(?)कालोपना तथा ।
शुद्धमध्य[ा] तथा चैव मार्गी स्यात्पौरवी तथा ॥ ६ ॥
दूष्यका(ती)वेति विज्ञेया सप्तमी द्विजसत्तमाः ।
मध्यमग्रामजा धत्येत्(?) विज्ञेयाः सप्त मूर्छनाः ॥ ७ ॥

आसी(सां) मध्यमगान्धारऋषभषड्जनिषादधैवतपञ्चमा आनुपूर्व्या आद्यग[ा]: । तत्र षड्जग्रामे—षड्जेनोत्तरमन्द्रा, निषादेन रजनी, धैवतेनोत्तरायता, पञ्चमेन शुद्धषड्जा, मध्यमेन मत्सरीत्यते(कृता), गान्धारेणाश्वक्रान्ता, ऋषभेणाभिरुद्धता, इति ।

अथ मध्यमसा(ग्रा)मे—मध्यमेन सौवीरी, गान्धारेण हरिणाश्वा, ऋषभेण कलोपनता, षड्जेन शुद्धमध्यमा, निषादेन मार्गी, धैवतेन पौरवी, पञ्चमेन दूष्यती, इति । एवमेताः क्रमयुक्ताः [पूर्ण]षाडवोडवितकृताः साधारणकृतास्य मूर्छनाश्चतुर्दश भवन्ति ।

अपि च—

क्रमयुक्ताः स्वराः सप्तमूर्छनास्वभिसंज्ञिताः ।
वस्वङ्कस्वरकास्थानाः षाडवोडविताः स्वराः ॥ ८ ॥
साधारणकृताश्चैव काकलीसमलंकृताः ।
प्र····ता स्वरसंयुक्ता मूर्छना ग्रामयोर्द्वयोः ॥ ९ ॥

natya1894, printed p.302, physical PDF p.305. Ṣaḍja count verses and definition of mūrchanā
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